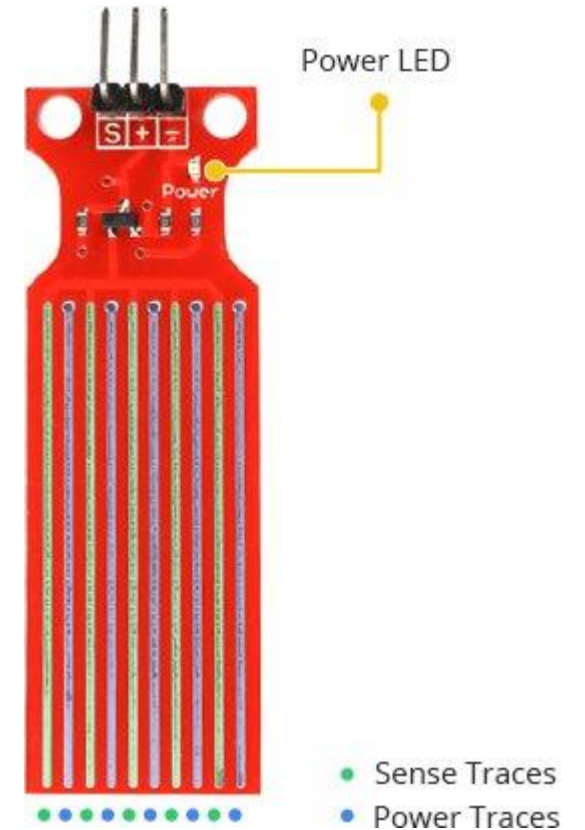


Examples of different types of sensors

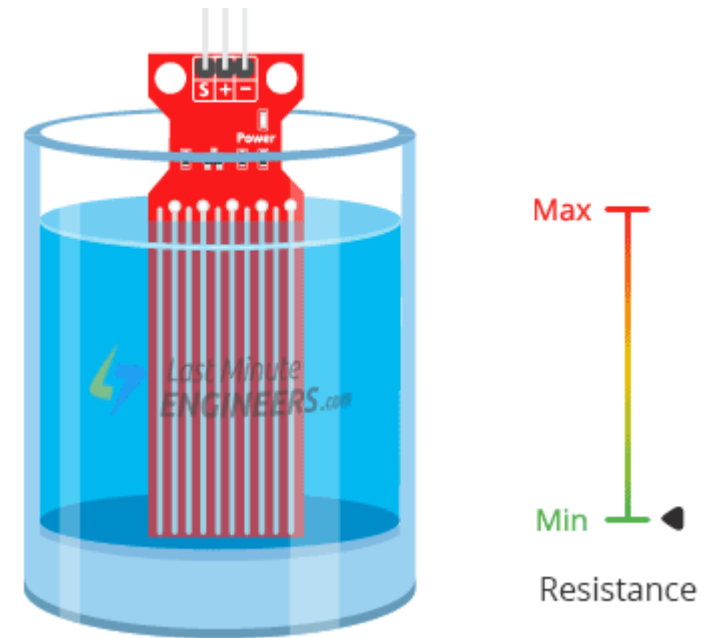
Water Level Sensor (Resistance)

- Presence of 10 exposed copper traces – 5 are power traces and 5 are sense traces
- One sense trace in between every two power traces
- These traces are bridged when submerged in water



Water Level Sensor working principle

- The parallel conductors act as a variable resistor and resistance varies according to water level
- Resistance inversely proportional to the height of the water
- More the immersion, better the conductivity and lower the resistance and vice versa
- Produces an output voltage according to resistance
- Issue: short lifespan as exposed to moist environment



Source: <https://lastminuteengineers.com/water-level-sensor-arduino-tutorial>

Optical water level sensor

infrared LEDs and phototransistors



Advantages: non-contact measurement, high accuracy, and fast response.

Disadvantages: Do not use under direct sunlight, water vapor will affect the measurement accuracy.

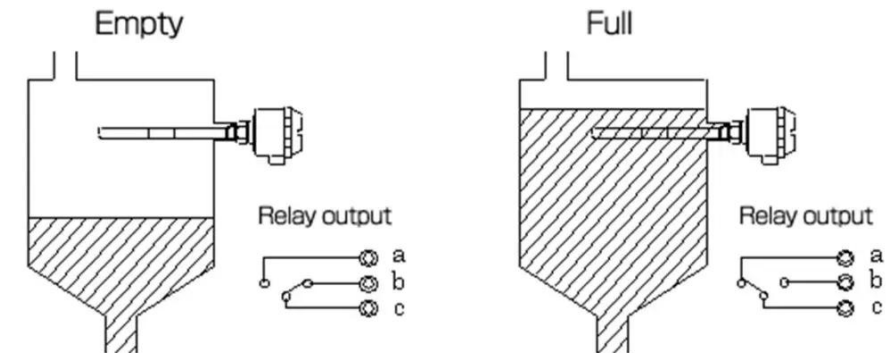
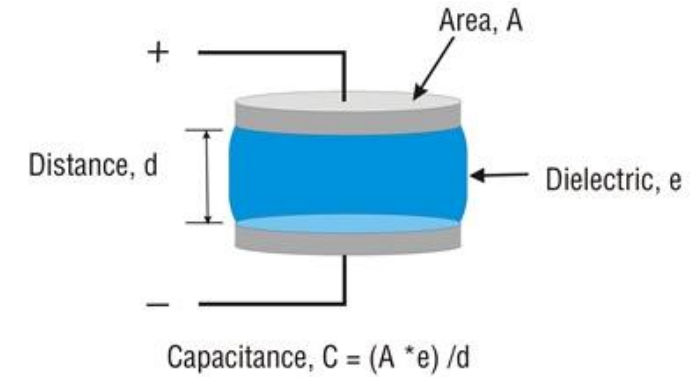
<https://www.renkeer.com/>

Capacitance liquid level sensor



Advantages: can be used to determine the rise or fall of the liquid in the container. By making the electrode and the container the same height, the capacitance between the electrodes can be measured. No capacitance means no liquid. A full capacitance represents a complete container.

Disadvantages: The corrosion of the electrode will change the capacitance of the electrode, and it needs to be cleaned or recalibrated.



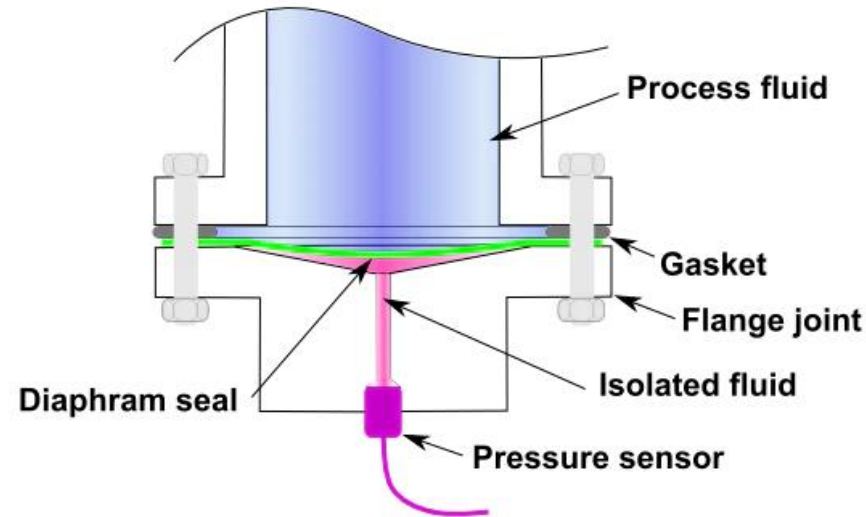
Tuning fork level sensor & Diaphragm liquid level sensor



resonant frequency.

Advantages: It can be truly unaffected by flow, bubbles, liquid types, etc., and no calibration is required.

Disadvantages: Cannot be used in viscous media.



Advantages: There is no need for power in the tank, it can be used with many types of liquids, and the switch will not come into contact with liquids.

Disadvantages: Since it is a mechanical device, it will need maintenance over time.

Float water level sensor & Ultrasonic liquid level sensor



Advantages: The float switch can measure any type of liquid and can be designed to operate without any power supply.

Disadvantages: They are larger than other types of switches, and because they are mechanical, they must be used more frequently than other level switches.

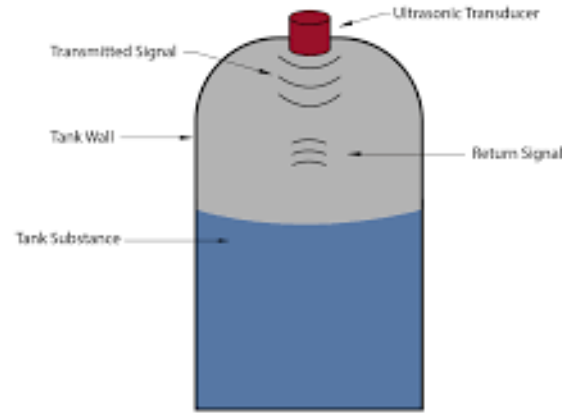


Figure 1: "Top-Down" Ultrasonic Sensor

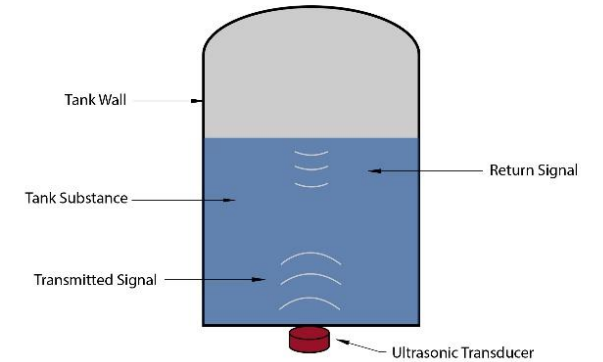
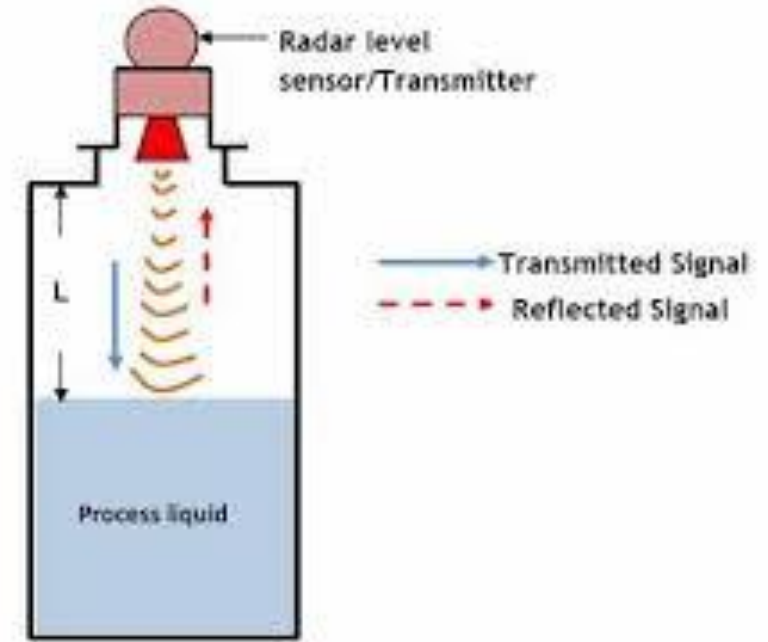


Figure 2: "Bottom-Up" Ultrasonic Sensor

Advantages: non-contact measurement, the measured medium is almost unlimited, and it can be widely used for measuring the height of various liquids and solid materials.

Disadvantages: The measurement accuracy is greatly affected by the temperature and dust of the current environment.

Radar level gauge



Advantages: wide application range, not affected by temperature, dust, steam, etc.

Disadvantages: It is easy to produce interference echo, which affects the measurement accuracy.

Ultrasonic

Natural Piezoelectric Materials



Quartz crystal
cluster from Tibet



Topaz



Sugar Cane



Tendon



DNA



Rochelle Salt



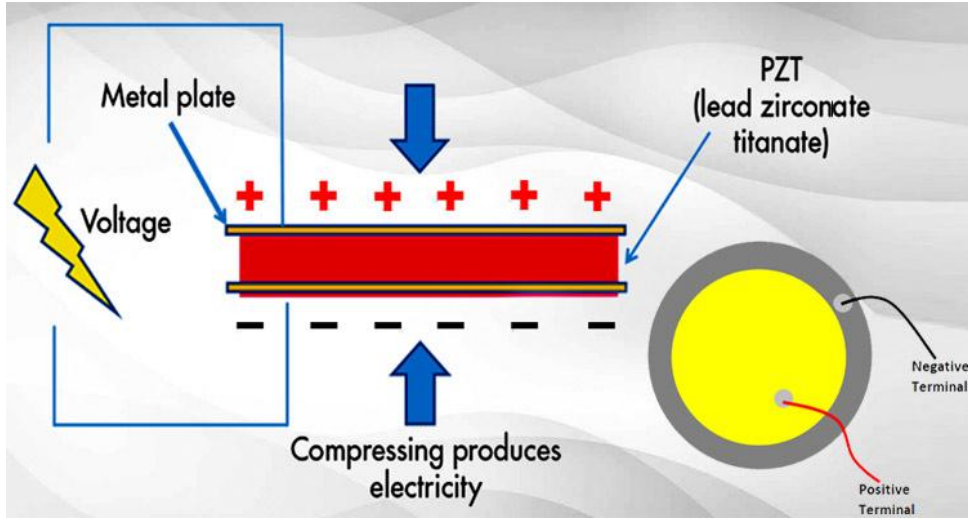
Schorl Tourmaline



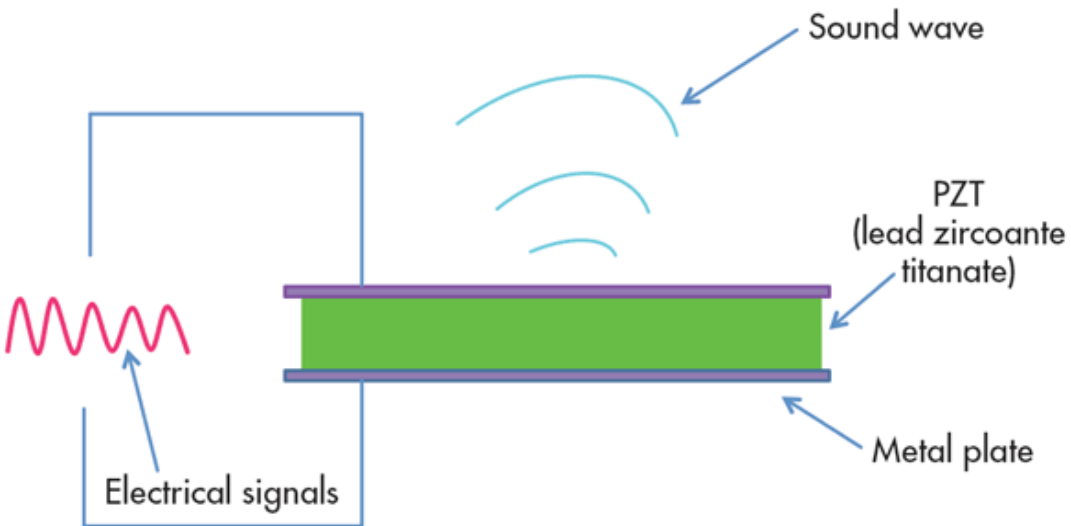
Dentine/ Enamel



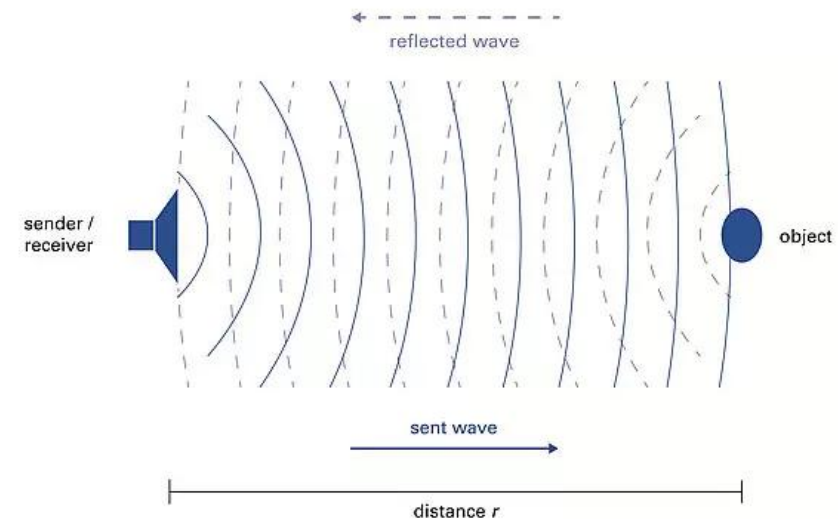
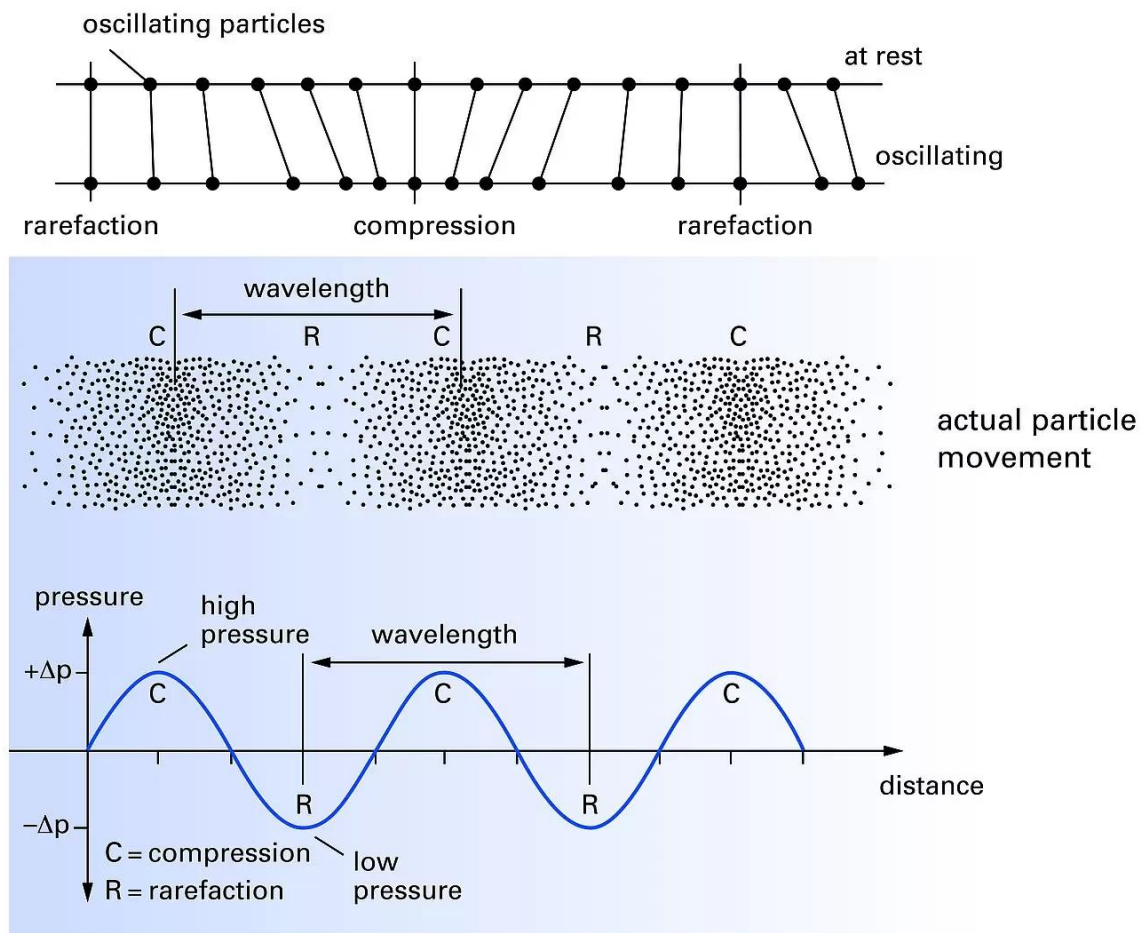
Bone



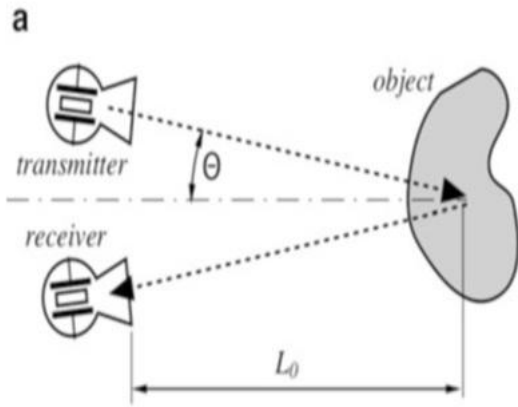
Applying mechanical energy to a crystal is called a direct piezoelectric effect which forces the excess negative and positive charges appear on opposite sides of the crystal face. The metal plate collects these charges, which can be used to produce a voltage and send an electrical current through a circuit.



Electrical energy applied to the crystal, shrinks and expands the crystal's structure. As the crystal's structure expands and contracts, it converts the received electrical energy and releases mechanical energy in the form of a sound wave.



Ultrasound - piezoelectric

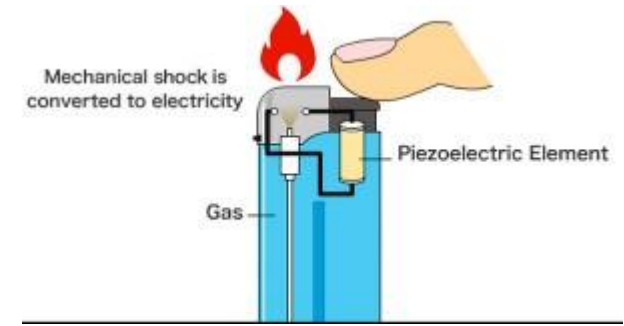


Moving object: Doppler effect

$$L_0 = vt \cos \theta / 2$$

<https://www.youtube.com/watch?v=y2pAFuSciB4&t=5s>

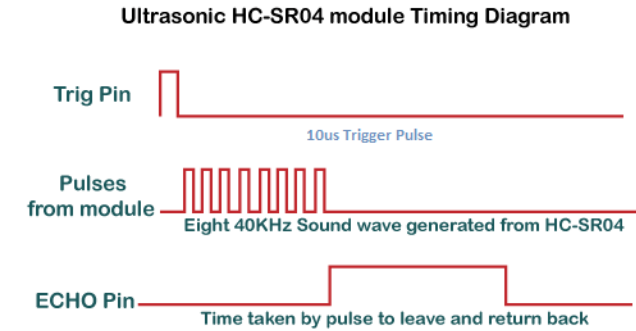
- Actuators use piezoelectricity to power devices - video cameras, and smartphones.
- Piezo sensors are used in a variety of applications such as microphones, amplified guitars, and medical imaging equipment.



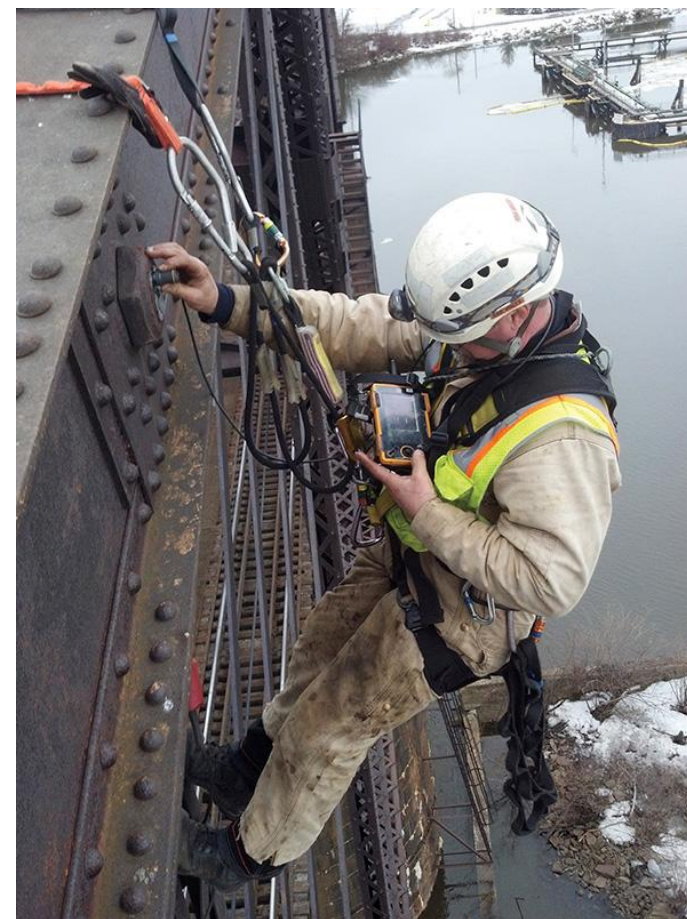
Pressing the button of the lighter releases a spring-loaded hammer into a piezoelectric crystal. This produces an electrical current that crosses a spark gap to heat and ignite gas

HC-SR04 Ultrasonic Sensor working principle

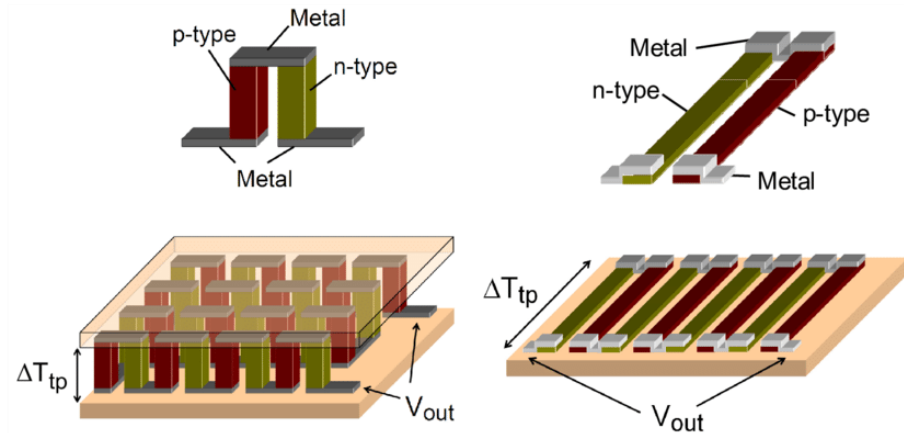
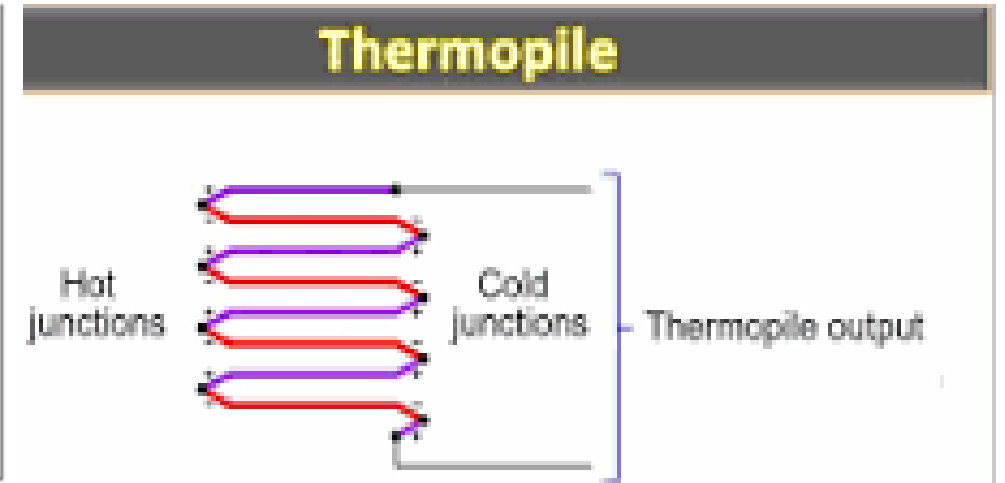
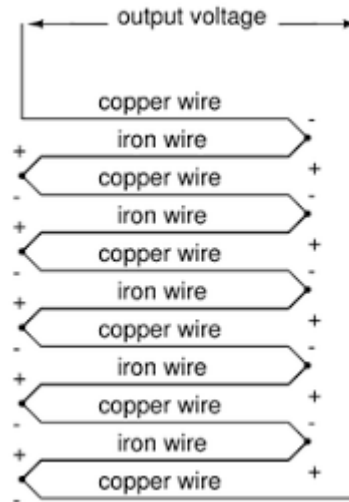
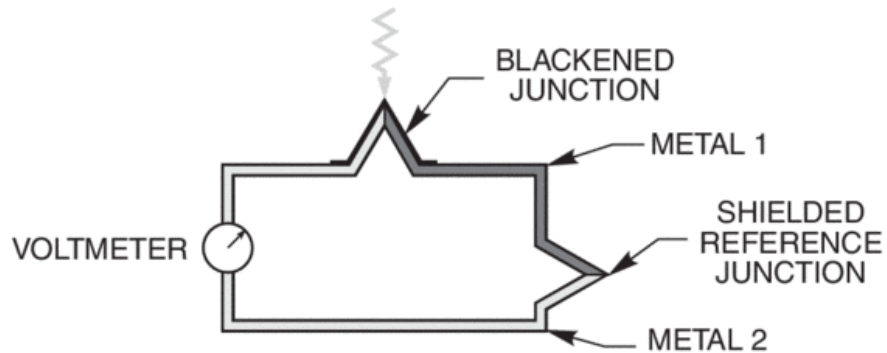
- Principle of the SONAR and RADAR system
- A pulse of at least 10 microsecond applied to Trigger pin
- Sensor transmits burst of 8 ultrasonic sound pulses at 40 KHz
- 8-pulse pattern makes an ultrasonic signature allowing receiver to differentiate from ambient ultrasonic noise
- As the ultrasonic pulses travel, the ECHO pin goes high
- If the pulses are not reflected, then ECHO signal will timeout in 38 msec and become low → indicates no obstruction or obstacle in the range
- If the pulses are received, the ECHO signal becomes low as soon as the signal is received
- A pulse produced whose width varies between 150 microseconds and 25 msec depending upon the time it took.



Other applications....



Temperature sensor -ZTP-135SR.



principle & Application

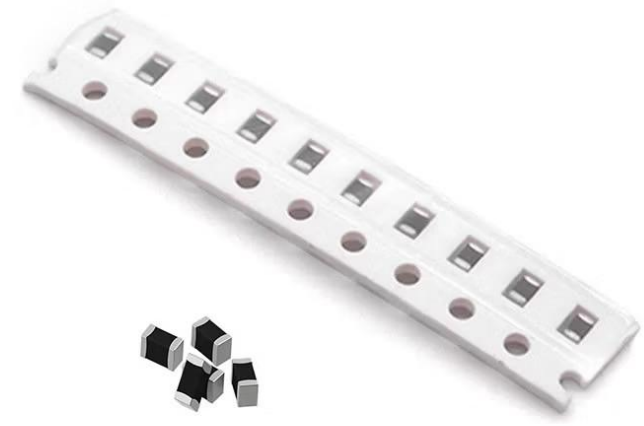
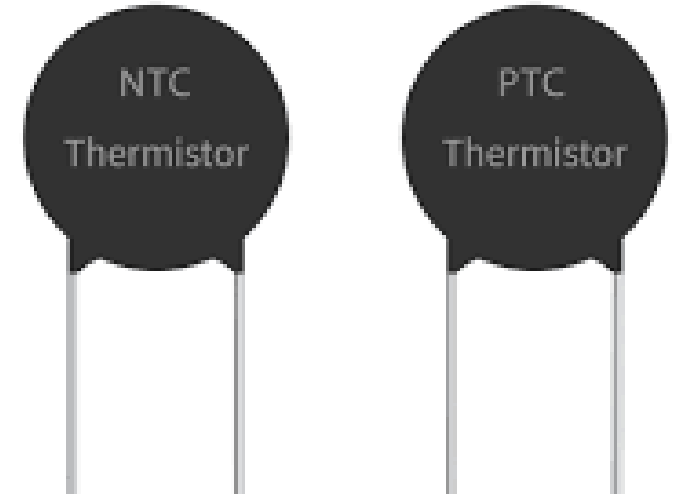
Range: -20 to 100 C
Response time: 25 ms



Thermistor

NTC & PTC

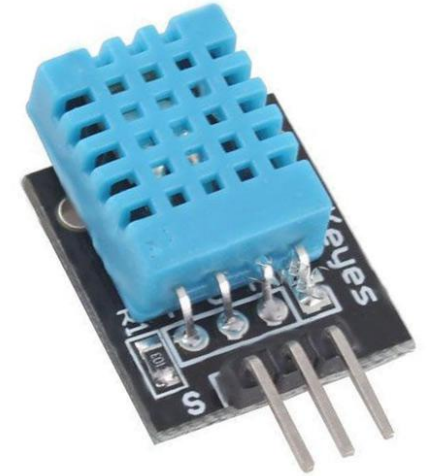
Surface-Mount Device thermistors – present in laptops & cellphone
leaded thermistors – mostly industry grade and used in thermal imagers



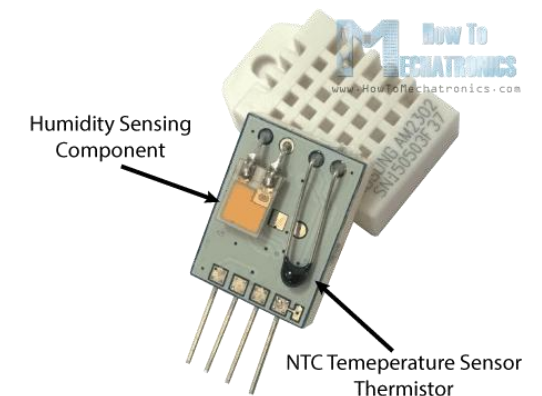
DHT11/22 sensor

- Basic low cost digital temperature and humidity sensor
- Uses capacitive humidity sensor and a thermistor to measure the surrounding air and outputs a digital signal on the data pin.
- A thermistor is a variable resistor that changes its resistance with changes in temperature
- Easy to use, can receive data once in every second
- Consists of a humidity sensing component, a NTC (Negative *Temperature* Coefficient) temperature sensor (thermistor) and an IC on the back side.

Source: <http://robocraft.ru/files/datasheet/DHT11.pdf>

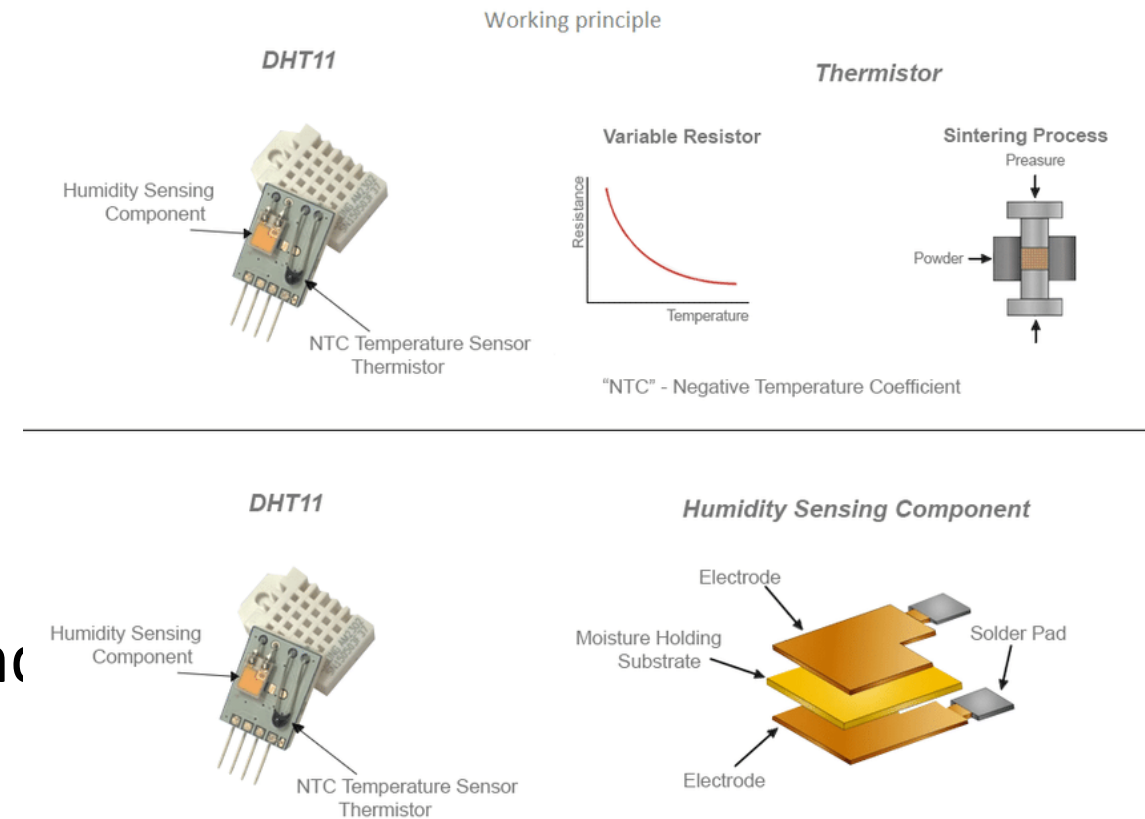


Source: <https://components101.com/dht11-temperature-sensor>



DHT11 working principle

- A humidity sensing component is used which has two electrodes and a moisture holding substrate between them
- Change in humidity → Conductivity of substrate changes or the resistance between the electrodes changes
- IC measures and processes this change and makes it ready to read



Source: <https://howtomechatronics.com/tutorials/arduino/dht11-dht22-sensors-temperature-and-humidity-tutorial-using-arduino/>

Passive IR Motion Sensor

- Every object/animal/human emits a certain amount of IR radiation
- PIR sensors can detect this IR radiation
- Use a pair of pyroelectric sensors to detect heat energy in the environment
- 2 sensors sit beside each other and when something enters the range, the differential between the sensors changes
- Sensors use a series of lenses which widen the sensing area



PIR materials

- Pyroelectricity is the effect in which some polar dielectric materials spontaneously change their polarization when their temperature is changed.
- Fundamentals: A polar material possesses a permanent dipole moment that is related to its molecule.
- If polar molecules are present then when an electric field is applied, the material will be polarized by orienting the dipole moments of polar molecules.
- If the P_s in a pyroelectric material can be switched by the external electric field, the material is called a ferroelectric (FE) material.
- Pyroelectric sensors: self-powered thermal sensors, gas alarms, fire alarms, pollution monitors, radiometers, laser detectors), imaging, electrochemistry (H_2 production, dye degradation, sterilization/disinfection), thin-film calorimeters, nanogenerators etc

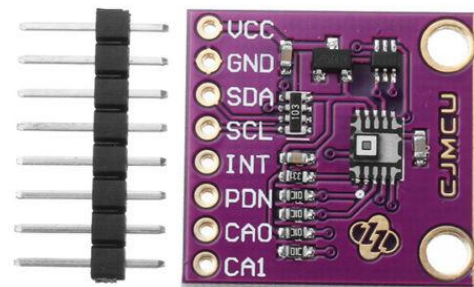
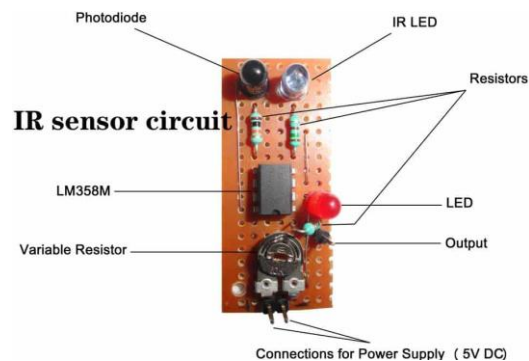
IR sensor

Motion Detection

Range: 100cm-500cm (1-5 meters / 3-15 feet)

Active: Active IR sensors have two parts: a light emitting diode (LED) and a receiver.

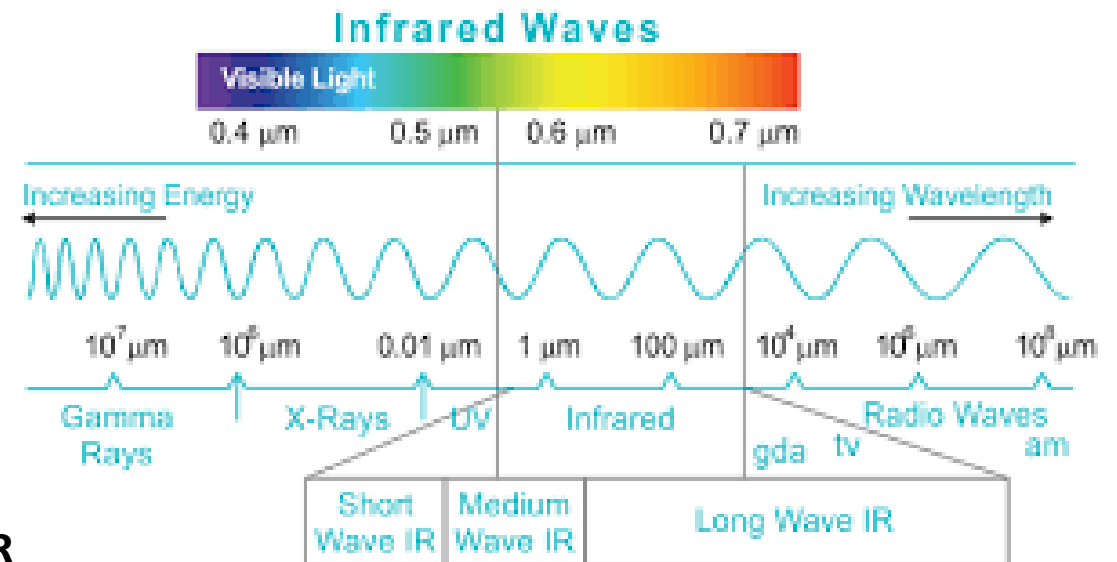
Passive. PIR?



CJMCU AK9750 Infrared Ray IR four quantum infrared human body detection sensor



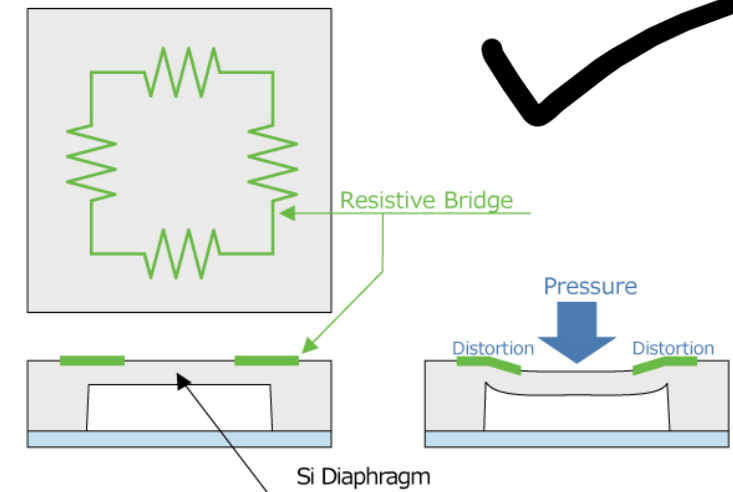
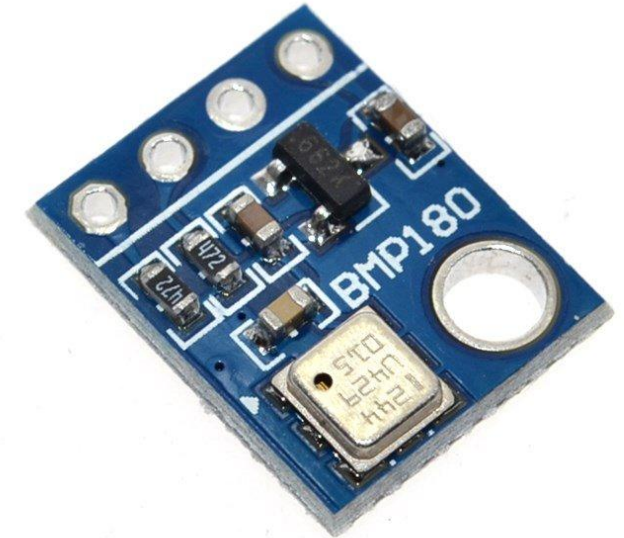
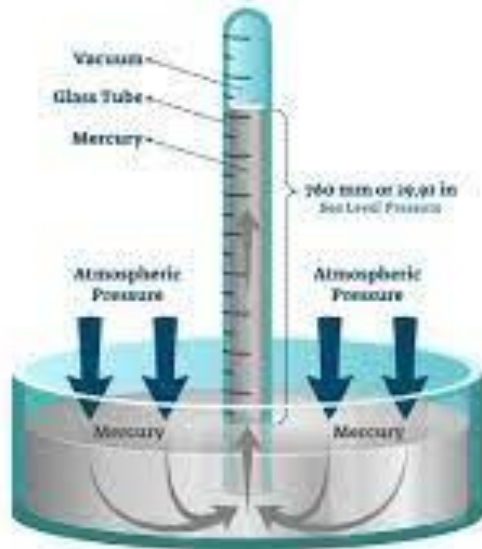
testbook



Digital Barometric Sensor (atmospheric)

- A piezoresistive sensor that detects pressure
- Measures both pressure and temperature

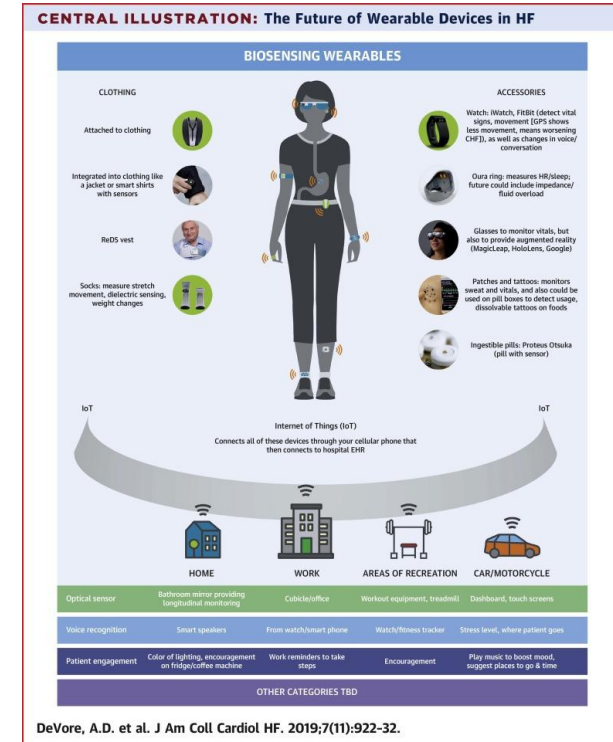
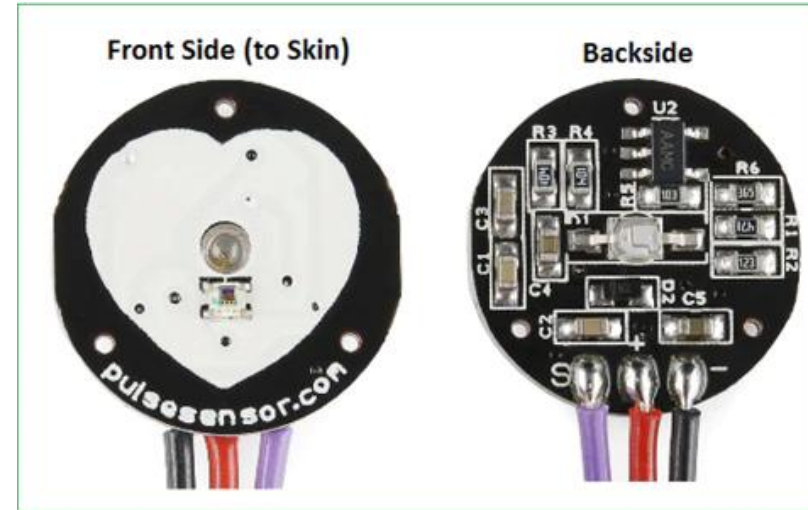
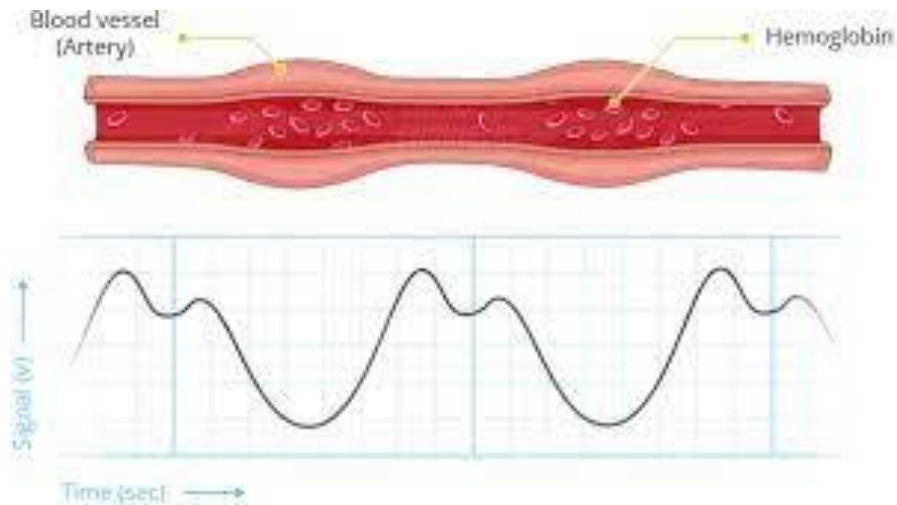
MERCURY BAROMETER



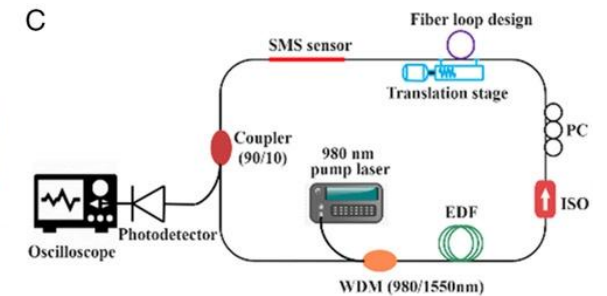
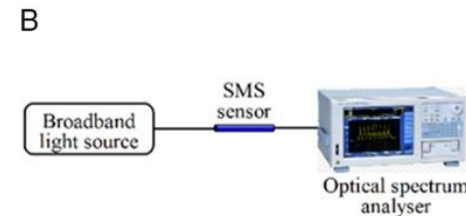
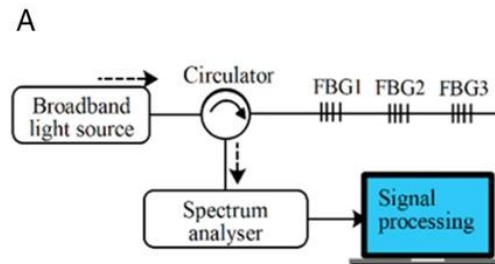
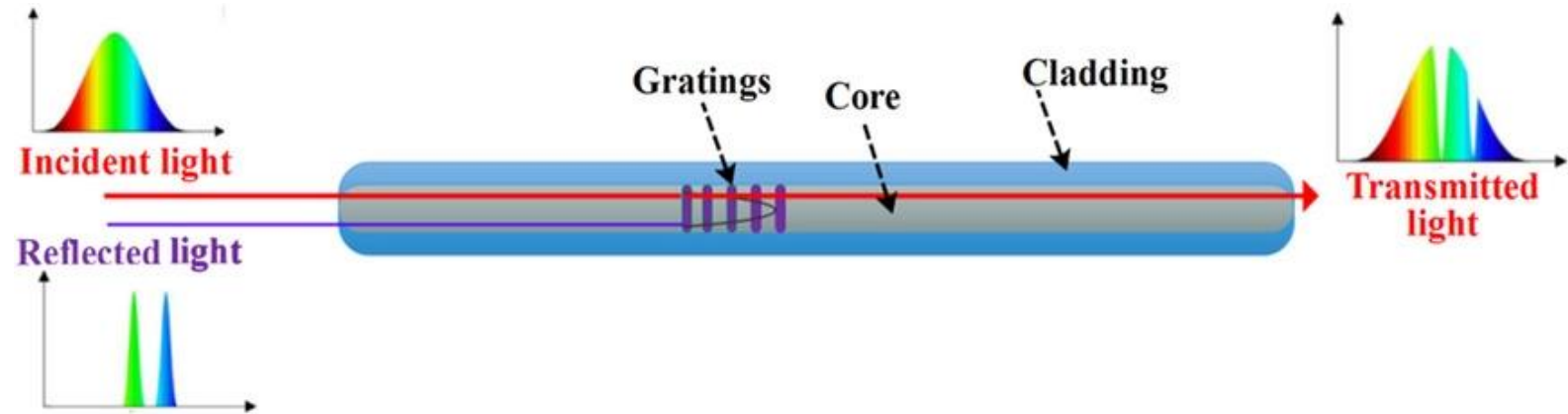
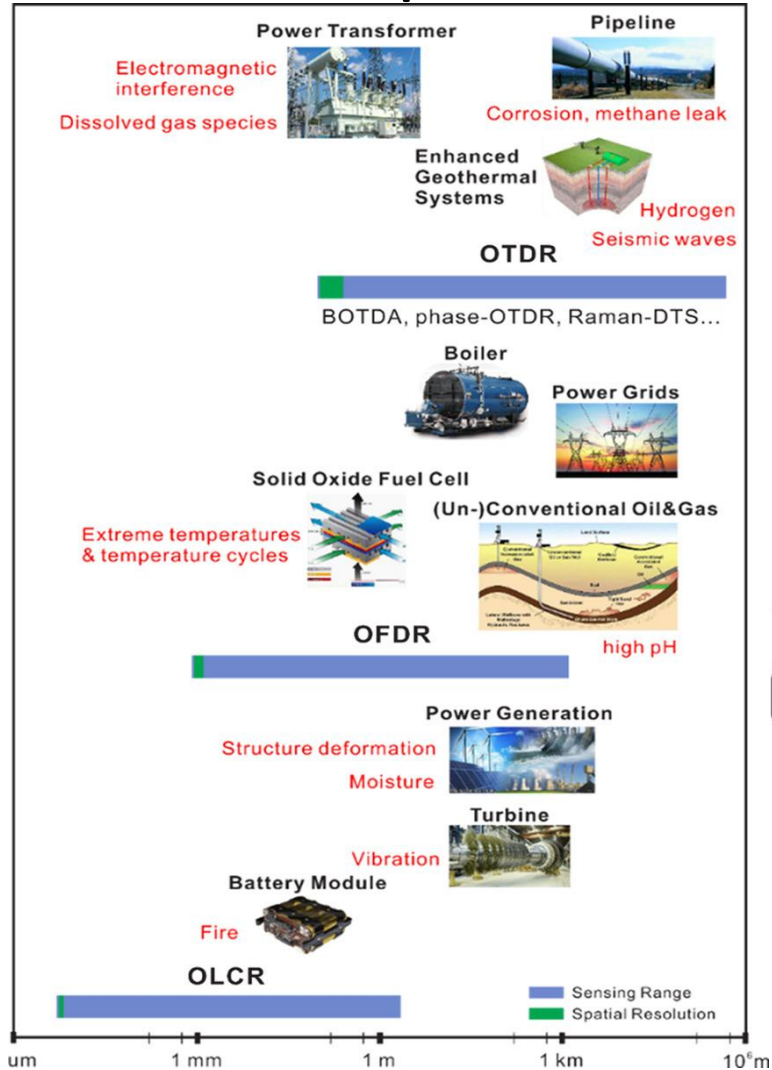
Pulse/HR sensor

LED: 550nm

Output data level: 5V TTL



Fiber Optic Sensors

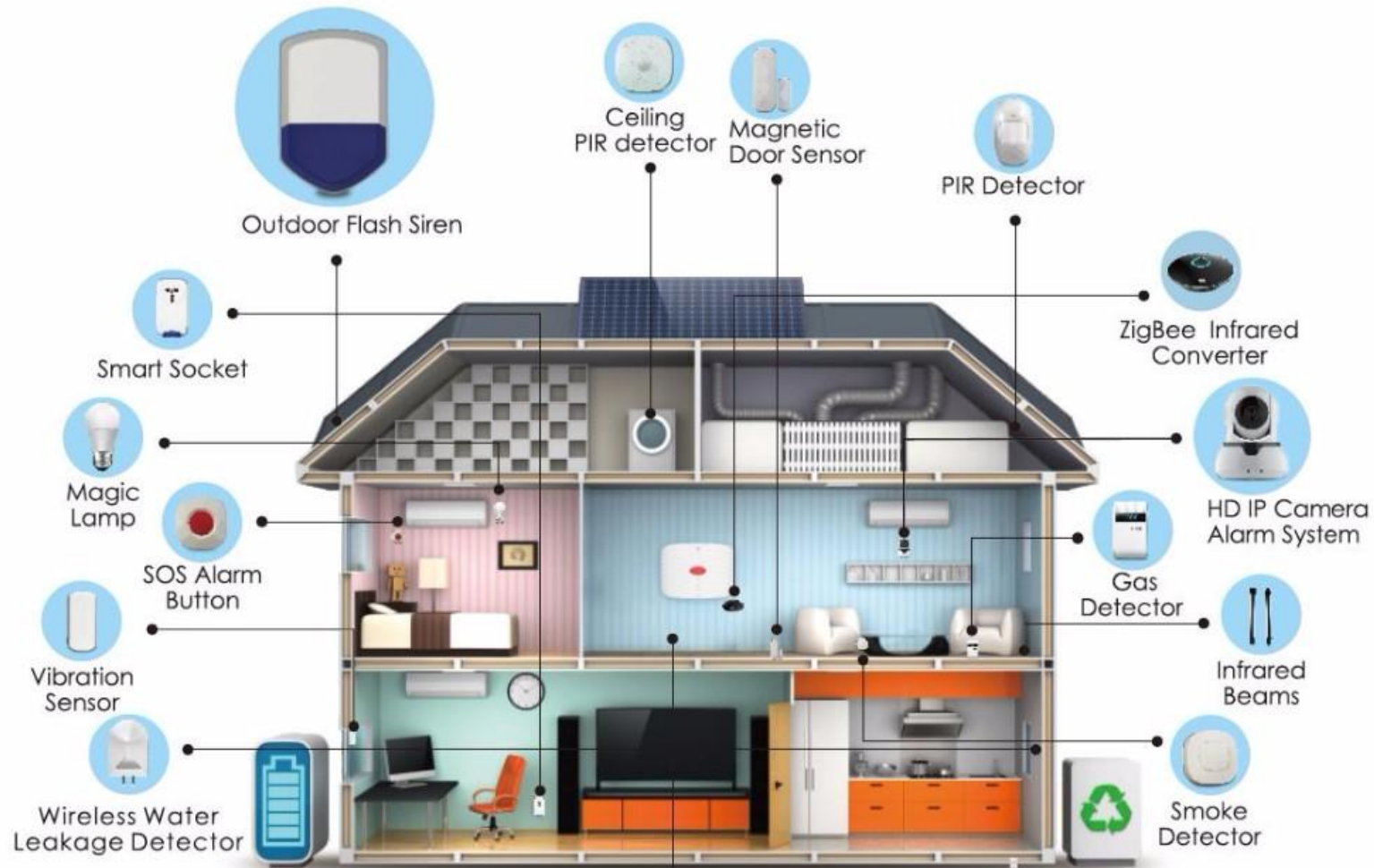


Other types

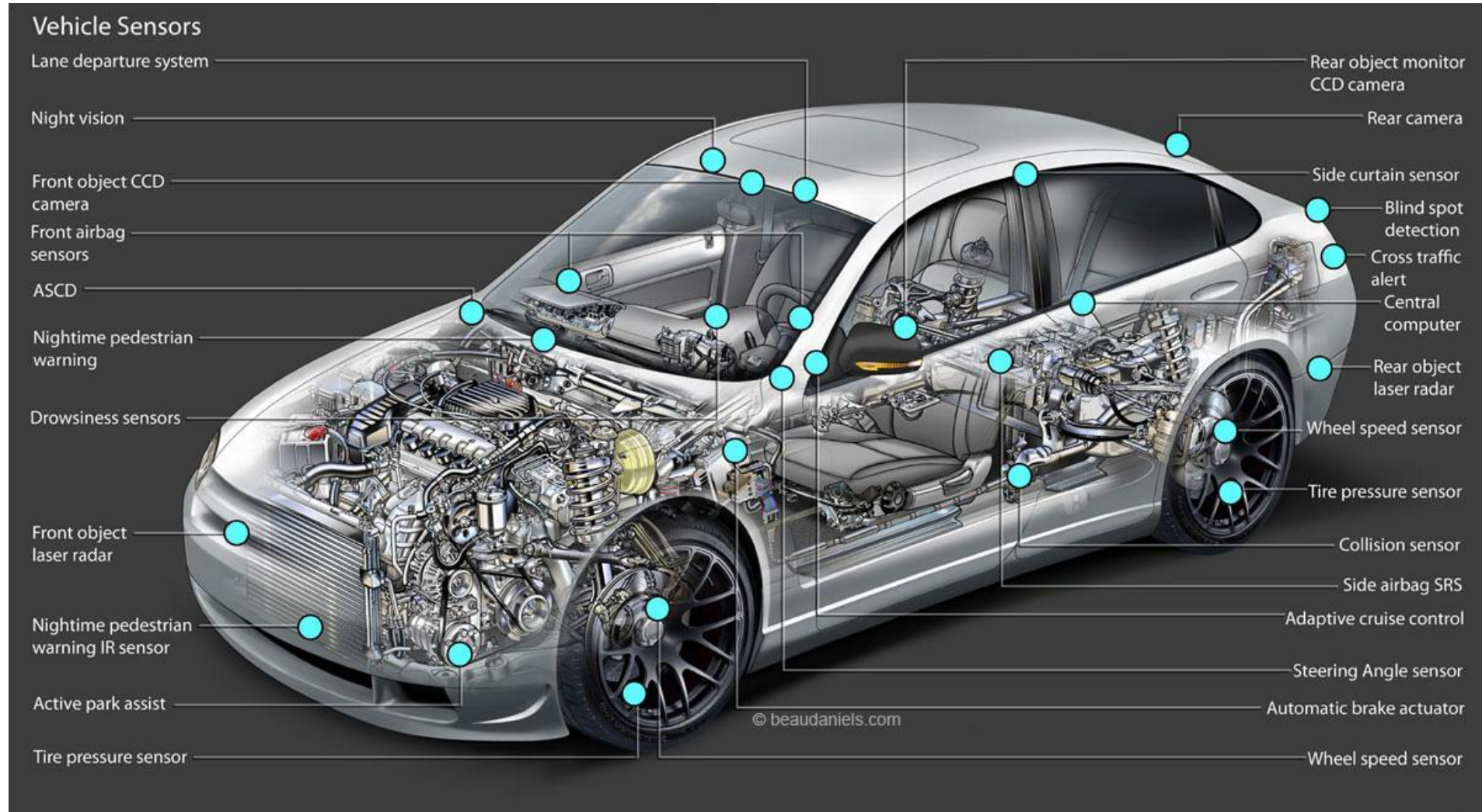
- MEMS (<https://www.bosch-mobility.com/en/solutions/electronic-components/mems-sensors/>)
- Biological sensors – use of bacteria or virus as the receptor organic material to detect.

Smart Home Sensors

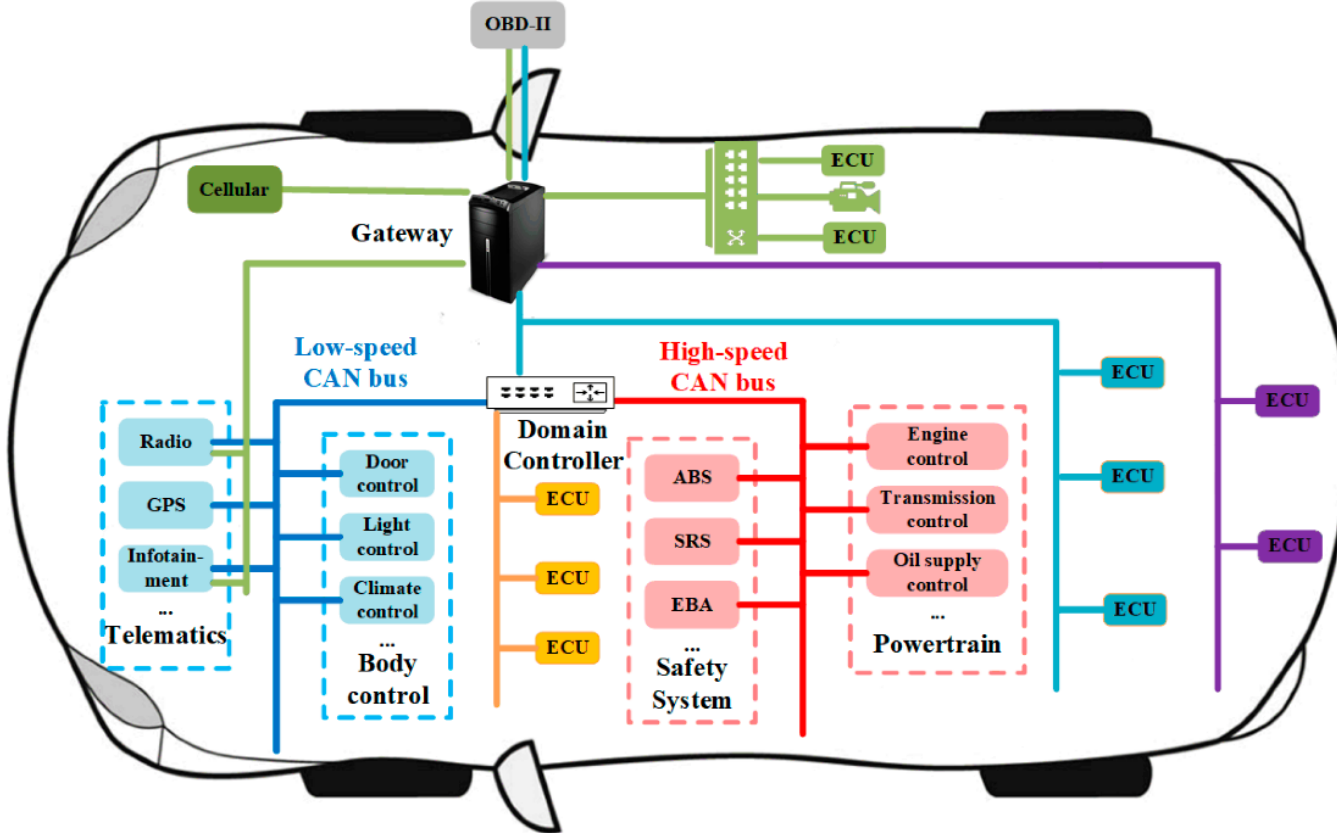
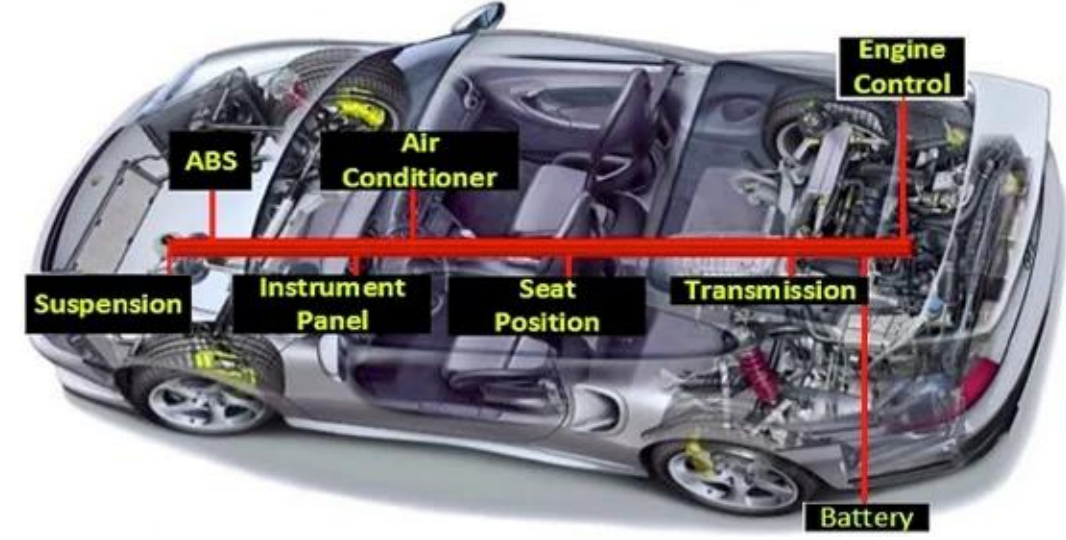
Smart Home



Sensors in a modern car



Controller area network



GPS: Global Position System
OBD-II: Second On-board Diagnostic

ABS: Anti-skid Brake System
SRS: Supplemental Restraint System
EBA: Emergency Brake Assist

— Ethernet — CAN-High
— FlexRay — LIN
— CAN — CAN-Low

Different Types of Sensors



IoT Hardware Platforms

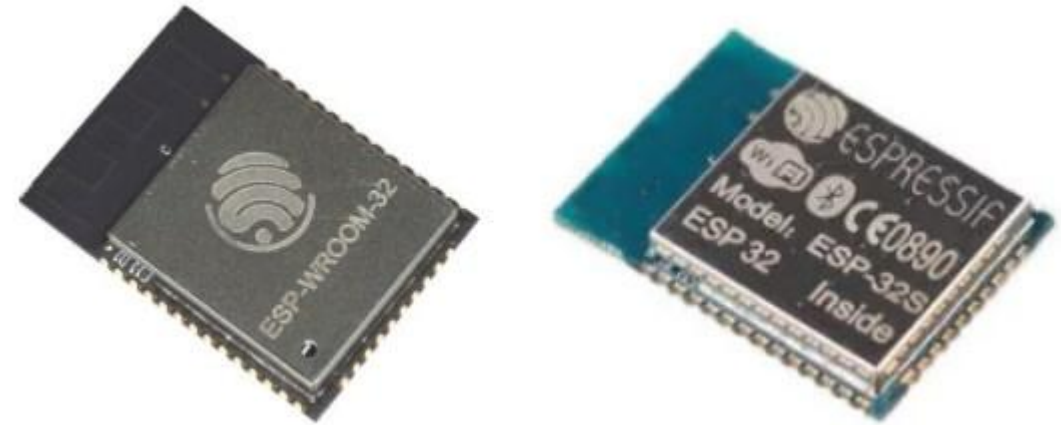
- Espressif ESP8266 Boards
 - Low cost, low power and easy to use
 - Initially used as a bridge to connect microcontrollers to wifi networks
 - Can be used as either a WiFi module connected to other microcontrollers or used in standalone mode
 - Small form factor



Source: <https://circuitdigest.com/article/top-hardware-platforms-for-internet-of-things-iot>

IoT Hardware Platforms

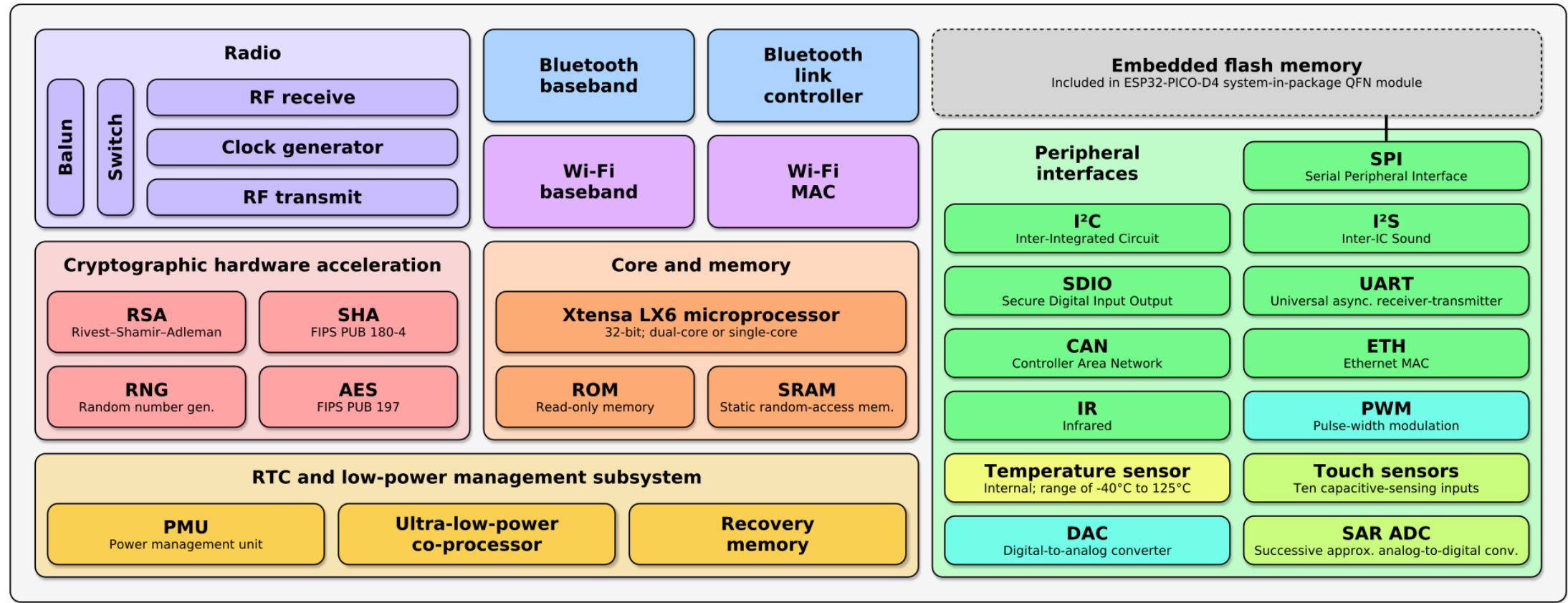
- ESP32 – Main features
 - Tensilica LX6 dual core processor at 240 MHz
 - 520 Kb of SRAM memory
 - WiFi 802.11 b/g/n connectivity with support for WEP, WPA/WPA2
 - Bluetooth connectivity (classic and LE)
 - 32 I/O pins with built in peripherals
 - Hardware acceleration for security algorithms



Source: <http://www.lucadentella.it/en/2016/12/03/esp32-1-introduzione/>

ESP32 functional block diagram

Espressif ESP32 Wi-Fi & Bluetooth Microcontroller — Function Block Diagram

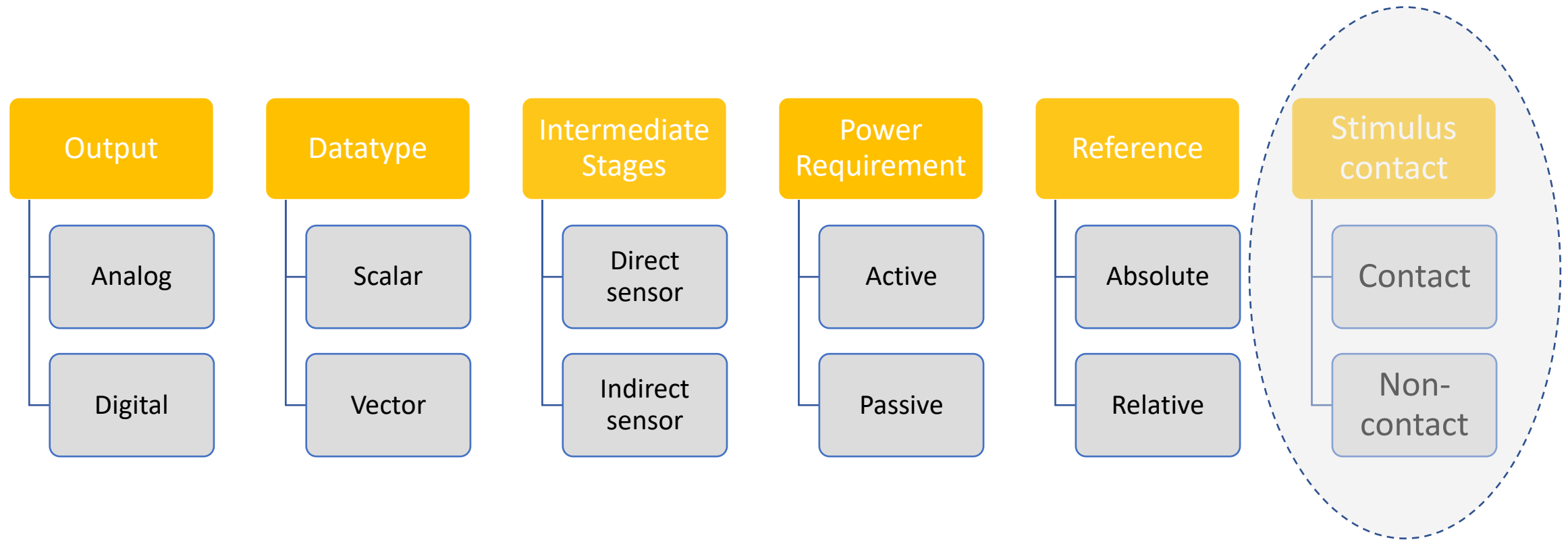


Summary – topics covered under sensors for IoT

Outline

- Definitions
- **Sensor Classification**
- **Sensor Characteristics**
- **Sensor Working Principles**

Types of Sensors



Sensor specifications

- Sensitivity
- Stimulus range (span)
- Stability (short and long term)
- Resolution
- Accuracy
- Selectivity
- Speed of response
- Environmental conditions
- Overload characteristics
- Linearity
- Hysteresis
- Dead band
- Operating life
- Output format
- Cost, size, weight Other

<https://www.signaguard.com/case-studies-in-bridge-health-monitoring/>

Conversion phenomena

Physical

- Thermoelectric
- Photoelectric
- Photomagnetic
- Magnetoelectric
- Electromagnetic
- Thermoelastic
- Electroelastic
- Thermomagnetic
- Thermo-optic
- Photoelastic
- Other

Chemical

- Chemical transformation
- Physical transformation
- Electrochemical process
- Spectroscopy
- Other

Biological

- Biochemical transformation
 - Physical transformation
 - Effect on test organism
 - Spectroscopy
- Other

Types of actuators

To enact movement, the actuator requires energy. The main types of energy sources are the following:

electric

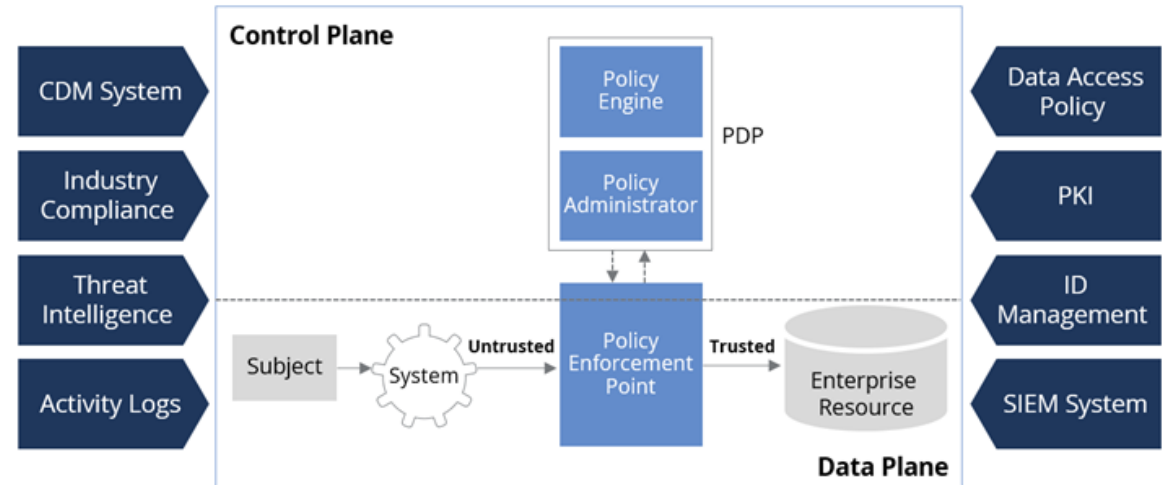
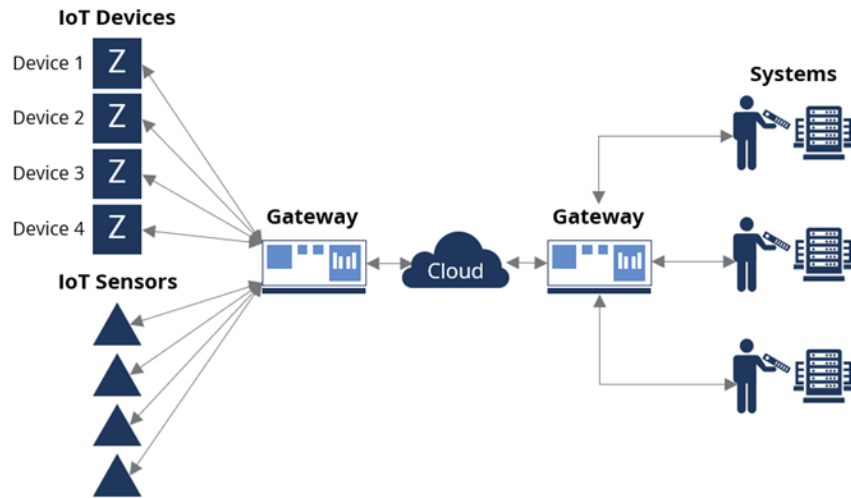
hydraulic

pneumatic

thermal/magnetic

Zero trust Architecture

Typical IoT environment with connections to an internal network



NIST 800-207 Zero Trust Framework